

CSE 370 – Database Systems

Assignment 02

Spring 2025

Submission Deadline : 22 April 2025 (Class Time)

Question 1 [CO5] : 5 Points

Consider the following relation:

T20CricketTournament (TournamentAcronym, Team_ID, Tournament_ID, Tournament_Title, Tournament_Matches, Base_Amount, Final_Amount, Team_Name, Team_Lead, Tournament_Year, Tournament_Sponsor, Tournament_Logo)

The primary key of the relation is underlined.

Suppose the following additional dependencies exist:

FD1: TournamentAcronym → Tournament_ID, Tournament_Title, Tournament_Matches, Base_Amount

FD2: Team_ID → Team_Name, Team_Lead

FD3: Tournament_ID → Tournament_Title, Tournament_Matches

Based on the given primary key,

- (i) Explain whether this relation is in 1NF. If not, decompose it to 1NF.
- (ii) Explain whether the relation of no (i) is in 2NF. If not, decompose it to 2NF.
- (iii) Explain whether the relation of no (ii) is in 3NF. If not, decompose it to 3NF.

Question 2 [CO5] : 5 Points

Consider the following relation:

CAR_SALE (Car, Salesperson, Commission, Date_sold, Discount_amt)

The primary key of the relation is underlined.

Suppose the following additional dependencies exist:

Suppose the following additional dependencies exist:

FD1 : Date_sold → Discount_amt

FD2 : Salesperson → Commission

Based on the given primary key,

- (i) Explain whether this relation is in 1NF. If not, decompose it to 1NF.
- (ii) Explain whether the relation of no (i) is in 2NF. If not, decompose it to 2NF.
- (iii) Explain whether the relation of no (ii) is in 3NF. If not, decompose it to 3NF.